Siddharth Deepak

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EDUCATION

B.S. in Business Analytics and AI | Concentration in Marketing and AI — Anticipated Graduation: Dec 2027

The University of Texas at Dallas – Naveen Jindal School of Management, Richardson, TX

RELEVANT COURSEWORK

Data Science for Business Applications, AI in Business, Business Analytics, Marketing Analytics, Foundations of Business Intelligence, Database Analytics, Object-Oriented Programming in Python, Quantitative Business Analysis

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, HTML/JS/CSS (familiar with R)

Libraries: Pandas, NumPy, scikit-learn, statsmodels, HistGradientBoosting, Optuna

Tools: Tableau, Power BI, Excel, Git, SQLite, Three.js, Chart.js, Firebase

Concepts: ML Modeling, Model Validation & Calibration, A/B Testing, Statistical Hypothesis Testing, Feature Ablation, Time-Series Forecasting, EDA, Data Cleaning

PROFESSIONAL EXPERIENCE

BizFalcon AI *bizfalconai.com* — Founder & Analytics Lead — May 2026 – Present

- Shipped two analytics products: **ProfitFalcon** for e-commerce brands — contribution margin, LTV/CAC, discount leakage and stockout risk — and **ClientFalcon** for marketing agencies, multi-client reporting built on an explainable account-health score; both run on standard platform exports (Shopify, Meta, Google Ads, GA4, Klaviyo)
- Built the shared analysis engine underneath, **Talon**, which runs Python over a client's entire dataset rather than a sample; authored **9 industry metric packs** and **11 guided statistical workflows** (hypothesis testing, regression, cohort, time-series forecasting) with the formula behind every figure exposed so any number can be audited
- Designed a recommendation layer that reads the data and names the next action — cut this budget, reorder this SKU, review this price — with **projected dollar impact**, human approval required before anything is applied, then scores each decision against the realized result so recommendation quality is measurable
- Engineered and operate the full stack solo — React/TypeScript, Firebase, in-browser Python (Pyodide), 9-model LLM integration, Stripe billing — with **400+ automated tests** and CI

YMCA Central Texas — Summer Camp Counselor — May 2021 – Aug 2025

- Led groups of 15–25 campers across four summers and trained incoming counselors through a structured shadowing program
- Awarded Counselor of the Week (5×) and Rookie Counselor of the Year for leadership and camper experience

PROJECTS

Retail Customer Intelligence Platform — *Live Dashboard*

Python | Scikit-learn | Pandas | RFM | Market-Basket Analysis

- End-to-end customer analytics on **802,712 real transactions** (41 countries, £17.45M): K-means RFM segmentation into 4 tiers; a random-forest churn model scored **out-of-sample at 0.796 ROC-AUC against a 55.3% base rate**; sized the retention campaign with a **precision@k curve** (top 100 contacts are 90% real churners) and a cost model showing contact capacity, not cost, is the binding constraint; Apriori market-basket rules (top pair lift 24×) powering a cross-sell engine

NBA Game Outcome Prediction Model — *GitHub*

Python | HistGradientBoosting | Optuna | SQLite | nba_api

- Trained a gradient-boosted classifier on **10,624 real NBA games** (9 seasons via nba_api) and validated on the fully held-out 2024-25 season: **66.4% accuracy against a 54.4% always-home baseline** (AUC 0.727, Brier 0.211), with isotonic-calibrated probabilities, TimeSeriesSplit, and 2020 bubble games down-weighted after measuring the home-court collapse
- Ran a **feature ablation** on the 71-feature set: Elo differential alone reaches AUC 0.710, so the other 70 add only **+0.015 AUC — under the ±0.021 spread across CV folds**, improving calibration but not ranking
- Re-tested on the **complete 2025-26 season** (1,230 unseen games, independent source): AUC held at **0.718** but did not beat a closed-form Elo formula (0.718 vs 0.718), winning only on calibration (**Brier 0.212 vs 0.215**) — the value is probability quality, not ranking

Olist E-Commerce Revenue & Sales Analysis — *Live Dashboard*

Python | SQL | Pandas | Tableau | Excel

- Analyzed **96,478 e-commerce orders** across 9 source tables, joined into a star schema and queried in SQLite with CTEs and window functions; Health & Beauty led revenue at R$1.4M of R$15.4M. At order grain, a logistic regression controlling for state, category and freight put a one-week delivery slip at **1.78× the odds of a ≤3-star review** (OR 1.085/day, 95% CI 1.083–1.088, n=95,824) — a larger effect than the raw 3.01★ vs 4.41★ gap implies. Published as a live Tableau dashboard.